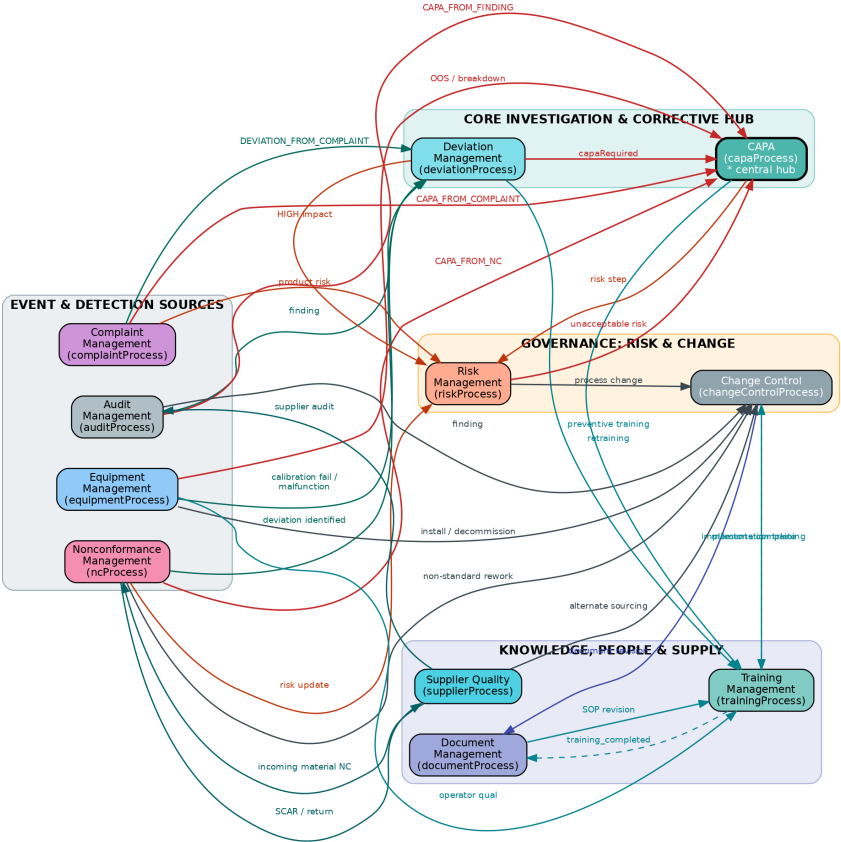


Mlabs QMS

Integrated Workflow Architecture & Cross-Module Integration Overview

How eleven quality modules connect — Flowable BPMN orchestration, task assignment, notifications and cross-module triggers



Master cross-module integration map (detailed on a later page)

Scope	11 QMS modules · 1 shared Flowable engine · 1 notification service · 40-table PostgreSQL schema
Source	Synthesized from the per-module sequence & workflow (.mmd) diagrams in this docs folder
Date	26 June 2026

1. Project Overview

The **Mlabs QMS** platform is an integrated, GxP-compliant electronic Quality Management System for a pharmaceutical manufacturing environment. It is built as a set of **eleven quality modules** that each manage one regulated quality process end-to-end, but which deliberately share a single technical backbone: one **Flowable BPMN 2.0 workflow engine**, one **Notification Service**, one electronic-signature and audit-trail framework, and one **PostgreSQL schema** (40 tables across 16 groups, including the shared *audit_trail*, *workflow_history*, *notifications*, *electronic_signatures* and *sequence_counters* tables). This shared backbone is what makes the modules behave as one system rather than eleven silos.

Every module is exposed through a **Spring Boot REST API** (/api/v1/<module>) backed by a module service (DeviationService, CapaService, ...). Each module owns one BPMN process definition (deviationProcess, capaProcess, ...) deployed to the same engine, so all human work surfaces in a single **unified Task Inbox** (GET /api/v1/tasks/inbox) regardless of which module created the task. The eleven modules are:

Module	BPMN process	Purpose
Deviation Management	deviationProcess	Captures unplanned departures from approved procedures.
CAPA	capaProcess	The corrective/preventive-action engine and central integration hub.
Change Control	changeControlProcess	Governs planned changes.
Complaint Management	complaintProcess	Handles customer/product complaints.
Nonconformance Management	ncProcess	Manages nonconforming material with optional immediate quarantine hold.
Audit Management	auditProcess	Plans and executes internal/external/supplier audits.
Risk Management	riskProcess	Maintains risk registers and assessments (FMEA/HACCP/FTA/PHA).
Supplier Quality Management	supplierProcess	Qualifies and monitors suppliers.
Training Management	trainingProcess	Assigns and verifies training.
Document Management	documentProcess	Controls SOPs and quality documents.
Equipment Management	equipmentProcess	Manages equipment lifecycle.

The remainder of this document describes the common architecture, how the Flowable engine is invoked and how tasks are assigned and notified, the cross-module connecting points that tie the modules together, and a compact flow for each module.

2. Common Architecture

All modules share the same five-layer runtime. A request flows top-down on submission and the engine then drives the process forward, calling back into the API, the notification service and the database as it advances.

Layer	Responsibility
1 · Presentation / Users	Role-based UI and a single unified Task Inbox. Users act as END_USER, QA_REVIEWER, QA_APPROVER, OWNER, AUDITOR, TRAINING_COORDINATOR, DOC_REVIEWER, CALIBRATION_TECH, etc.
2 · Application (Spring Boot REST API)	One service per module exposing POST /api/v1/<module> to create records and PATCH /{id}/{action} to advance them. Generates human-readable sequence numbers (DEV-2026-001...), enforces validateClosureRules() and e-signature checks.
3 · Workflow Engine (Flowable BPMN 2.0)	RuntimeService starts process instances; TaskService manages claim/complete; exclusive gateways route on process variables; JavaDelegate service tasks (notification/escalation/cross-module) and boundary timer events automate the process.
4 · Notification Service	Generates in-app notifications (types TASK_ASSIGNED, STATUS_CHANGE, ESCALATION, *_DUE) when fired by Flowable service tasks, and persists them to the shared <i>notifications</i> table.
5 · Persistence (PostgreSQL)	40 tables / 16 groups. Module tables hold records and cross-module foreign keys; shared tables provide audit trail, workflow history, notifications, e-signatures and sequence counters. Polymorphic record_type + record_id links any shared row to any quality record.

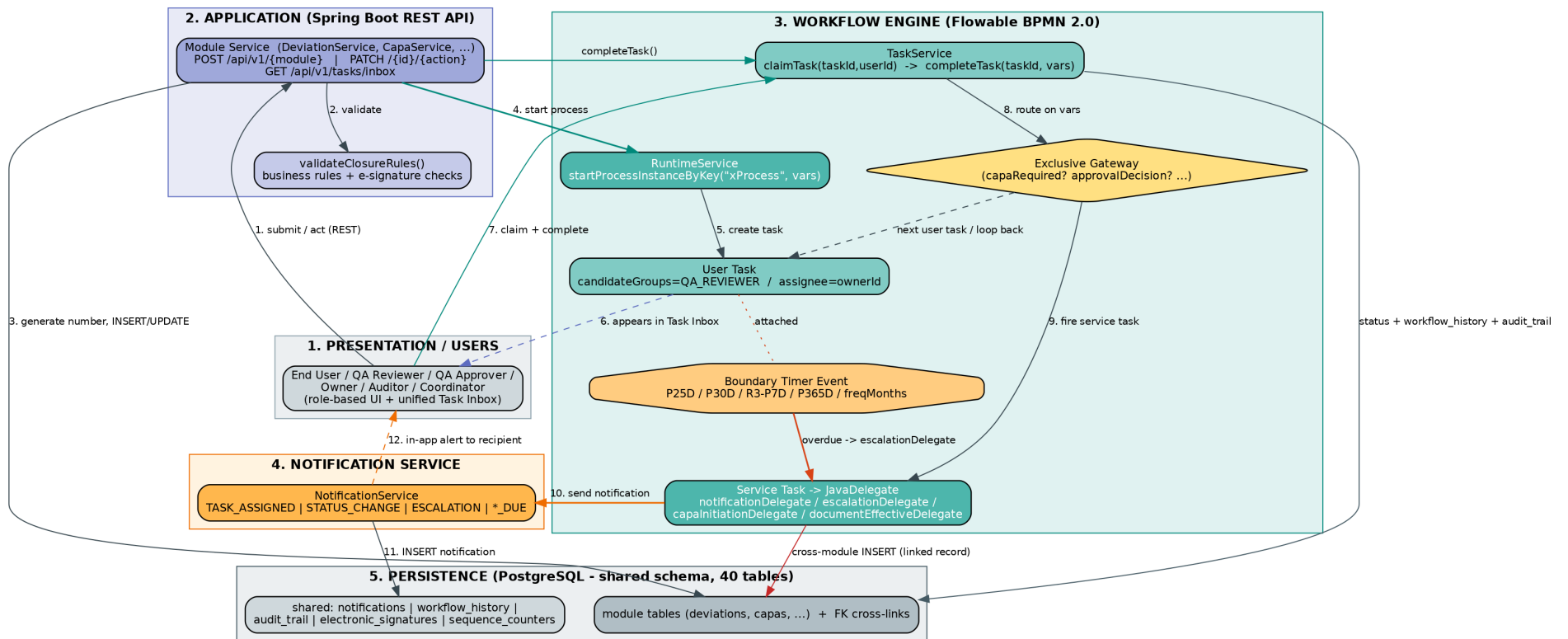
3. How the Flowable Engine is Invoked

Every module follows the same orchestration pattern. The numbered sequence below corresponds to the backbone diagram on the next (landscape) page:

- 1. Submit (REST).** A user POSTs a new record (or PATCHes an action) to the module service.
- 2-3. Persist.** The service validates, generates a sequence number and INSERTs/UPDATEs the record in PostgreSQL.
- 4. Start process.** The service calls `runtimeService.startProcessInstanceByKey("<module>Process", vars)`, passing record id, owner, dates and decision variables.
- 5. Create task.** The engine reaches the first *user task* and creates a work item assigned either to a **candidate group** (e.g. `candidateGroups=QA_REVIEWER`) or to a specific **assignee** (e.g. `assignee=ownerId`).
- 6. Surface in inbox.** The task appears for all eligible users via `GET /api/v1/tasks/inbox`.
- 7. Claim & act.** A user `claimTask(taskId, userId)` to take ownership and completes the work in the UI form.
- 8. Complete.** The service calls `completeTask(taskId, vars)`, returning decision variables such as `capaRequired`, `approvalDecision` or `reviewDecision`.
- 9. Route.** An *exclusive gateway* evaluates those variables and routes to the next user task, a service task, a revision loop, or an end event.
- 10. Automate.** *Service tasks* execute `JavaDelegates` — for notifications, escalations or to create a linked record in another module.
- 11-12. Notify.** The notification delegate writes a row to *notifications* and pushes an in-app alert to the recipient. In parallel, *status*, *workflow_history* (COMPLETED / CURRENT / PENDING) and *audit_trail* rows are written.

Revision loops (document author revision, CAPA effectiveness failure, audit response revision, training fail-retry) simply route the gateway back to an earlier user task until the exit condition is met. The process ends at a normal, rejected or voided end event.

Flowable Invocation & Notification Backbone (shared by all 11 modules)



Numbered steps 1-12 match Section 3. Red = cross-module record creation; orange = notification path; dark-red octagon = boundary-timer escalation.

4. Task Assignment & Notification Model

Assignment. Tasks are assigned in one of two ways. **Candidate-group** tasks (`flowable:candidateGroups`) are offered to every user holding a role — review, effectiveness and approval steps typically go to QA_REVIEWER or QA_APPROVER groups, and a user claims the task to take it. **Direct-assignee** tasks (`flowable:assignee`) go to one named person — investigation, action execution and implementation go to the record owner or a named investigator/trainee. The unified inbox merges both across all eleven processes.

Notification. Notifications are never sent ad-hoc from controller code; they are emitted by Flowable **service tasks** that invoke a `JavaDelegate`. Two delegates dominate:

- `notificationDelegate` — routine in-app notifications at workflow milestones: a task being assigned (`TASK_ASSIGNED`), a status change such as a closure (`STATUS_CHANGE`), and due-date reminders (e.g. `CALIBRATION_DUE`, `PERIODIC_REVIEW_DUE`, `RETRAINING_DUE`).
- `escalationDelegate` — overdue escalations fired by boundary timers (`INVESTIGATION_OVERDUE`, `IMPLEMENTATION_OVERDUE`, `AUDIT_REPORT_OVERDUE`, `TRAINING_OVERDUE`, `REGULATORY_REPORTING_OVERDUE`), typically notifying both the owner and a supervisor/coordinator.

Timers. Two timer styles are used. **Boundary timers on active tasks** fire when a step runs late (e.g. P25D on deviation investigation, P30D on change implementation, R3/P7D three reminders on CAPA actions). **Periodic timers** on completed/effective records drive recurring lifecycle events (calibration, maintenance, re-qualification, retraining and periodic document/risk review). All are non-interrupting, so the reminder or escalation fires without cancelling the underlying work.

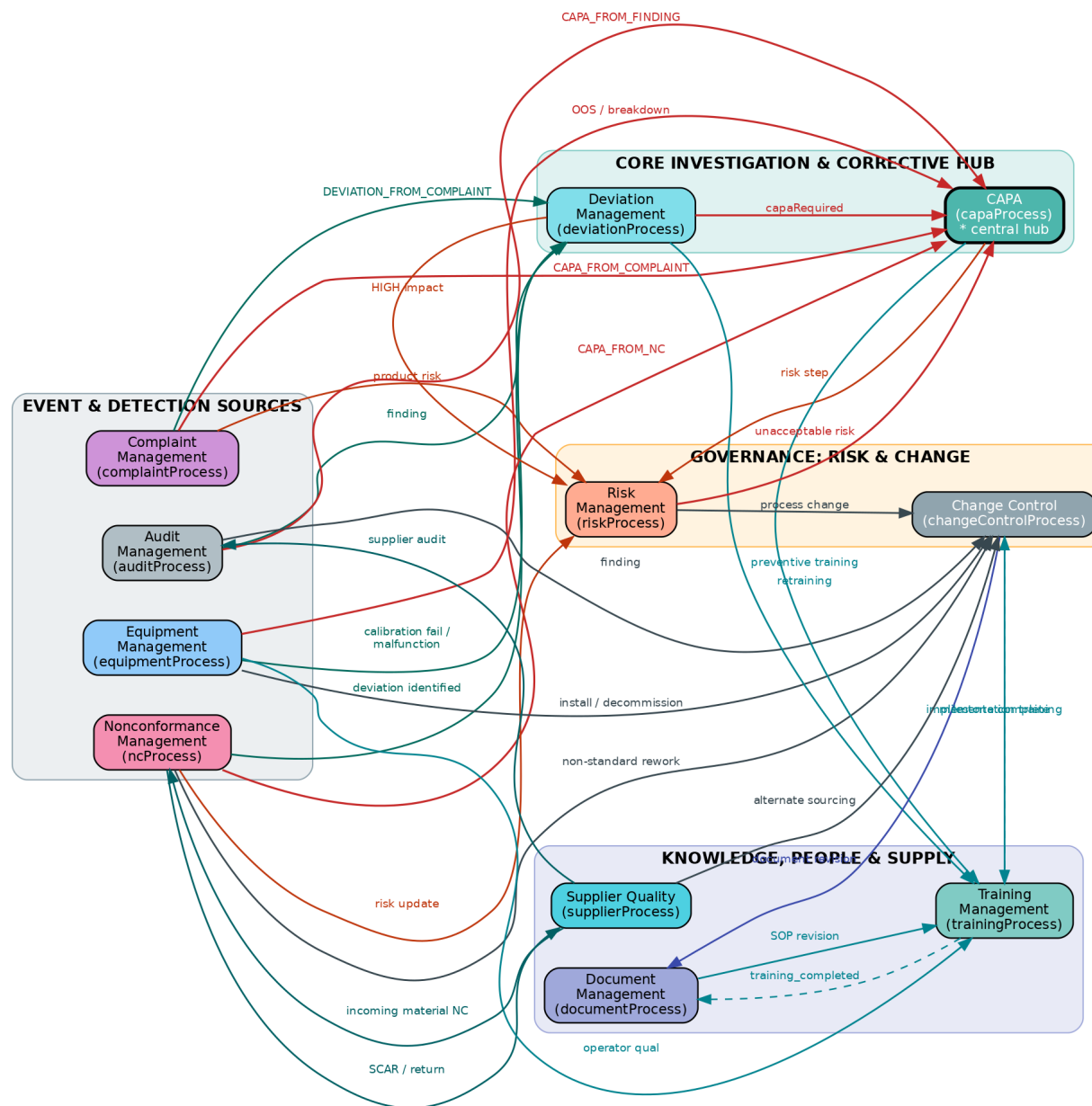
Compliance. Every state transition records an *audit_trail* entry and a *workflow_history* step (`COMPLETED` / `CURRENT` / `PENDING`), and disposition/approval/closure steps require an **electronic signature** — together providing the 21 CFR Part 11 traceability the regulated environment requires.

5. Cross-Module Integration & Connecting Points

The connecting points between modules are themselves implemented as Flowable **service-task delegates**. When one process reaches a decision that requires action in another domain, a delegate (`capaInitiationDelegate`, `deviationDelegate`, `changeInitiationDelegate`, `auditDelegate`, `trainingDelegate`, `sourceUpdateDelegate`) INSERTs a new record in the target module with a **foreign-key link and a sourceType** (e.g. `CAPA_FROM_NC`, `sourceType=AUDIT_FINDING`), then starts that module's own BPMN process. The new record runs its full independent lifecycle while remaining traceable back to its origin.

CAPA is the central hub — Deviation, Complaint, Nonconformance, Audit, Equipment and Risk can all spawn a CAPA. **Risk, Change Control and Training are secondary hubs:** Risk aggregates risk signals from five modules; Change Control receives change needs from Risk, Audit, Nonconformance, Supplier and Equipment and then propagates document revisions and training; Training is fed by Document, Change, CAPA, Deviation and Equipment and back-updates each source on completion. The full picture:

Master Cross-Module Integration Map



Arrow colour groups the target: red = CAPA, green = Deviation, orange = Risk, dark = Change Control, blue = Document, teal = Training/Supplier. Dashed = back-update.

6. Cross-Module Trigger Matrix

Each row is a connecting point: the condition in the source process that fires a delegate to create a linked record (and start the process) in the target module.

Source	Trigger condition	Target	Mechanism
Deviation	Disposition: capaRequired = true	CAPA	capaInitiationDelegate
Deviation	HIGH impact assessment	Risk	linkedDeviation FK
Deviation	Closure requires retraining	Training	trainingDelegate
CAPA	Risk-assessment step	Risk	linkedCapa FK
CAPA	Preventive action = training	Training	trainingDelegate
Complaint	Disposition: capaRequired	CAPA	capaInitiationDelegate (sourceType COMPLAINT)
Complaint	Deviation identified	Deviation	deviationDelegate
Complaint	Product risk pattern	Risk	linkedComplaintId
Nonconformance	capaRequired = true	CAPA	capaInitiationDelegate (sourceType NONCONFORMANCE)
Nonconformance	Deviation identified	Deviation	deviationDelegate
Nonconformance	supplierId set	Supplier	SCAR / material return, score update
Nonconformance	Non-standard rework	Change Control	changeInitiationDelegate
Nonconformance	Risk update required	Risk	update risk register
Audit	Finding: capaRequired	CAPA	capaInitiationDelegate (sourceType AUDIT_FINDING)
Audit	Finding needs deviation	Deviation	deviationDelegate
Audit	Finding needs change	Change Control	changeInitiationDelegate
Supplier	Doc review AUDIT_REQUIRED / re-qual	Audit	auditDelegate (auditType SUPPLIER)
Supplier	Incoming material issue	Nonconformance	ncType MATERIAL, supplier FK
Supplier	Disqualification	Change Control	changeInitiationDelegate (alternate sourcing)
Change Control	Implementation: document affected	Document	document revision
Change Control	Implementation training	Training	trainingDelegate
Change Control	Impact assessment	Risk	linked_change_id
Risk	Process change required	Change Control	changeInitiationDelegate (sourceType RISK_ASSESSMENT)
Document	Document effective (SOP revision)	Training	training_assignments (reason SOP_REVISION)
Training	Verification complete	Source module	sourceUpdateDelegate (training_completed)
Equipment	GxP install / decommission	Change Control	changeInitiationDelegate
Equipment	Qualification needs operators	Training	trainingDelegate
Equipment	Calibration fail / breakdown	Deviation	deviationDelegate (+ CAPA if required)

7. Escalation, Timer & Periodic-Review Catalog

Boundary and periodic timers configured across the modules. Boundary timers escalate overdue work; periodic timers drive recurring quality obligations.

Module	Attached to	Timer	Event raised	Delegate
Deviation	Investigation	P25D	INVESTIGATION_OVERDUE	escalationDelegate
CAPA	Action execution	R3 / P7D (3×)	Overdue reminder	reminderDelegate
Change Control	Implementation	P30D	IMPLEMENTATION_OVERDUE	escalationDelegate
Complaint	Regulatory report	P15D	REGULATORY_REPORTING_OVERDUE	escalationDelegate
Complaint	Investigation	P20D	INVESTIGATION_OVERDUE	escalationDelegate
Audit	Conduct / report	reportDueDate	AUDIT_REPORT_OVERDUE	escalationDelegate
Training	Completion	dueDate	TRAINING_OVERDUE	escalationDelegate
Document	Effective record	P365D	PERIODIC_REVIEW_DUE	notificationDelegate
Risk	Approved register	reviewFrequencyMonths	RISK_REVIEW_DUE	notificationDelegate
Supplier	Qualified supplier	requalificationFrequencyMonths	REQUALIFICATION_DUE	notificationDelegate
Training	Verified record	validityMonths	RETRAINING_DUE	notificationDelegate
Equipment	Operational	calibrationFrequencyDays	CALIBRATION_DUE	notificationDelegate
Equipment	Operational	maintenanceFrequencyDays	MAINTENANCE_DUE	notificationDelegate
Equipment	Operational	12 months	REQUALIFICATION_DUE	notificationDelegate

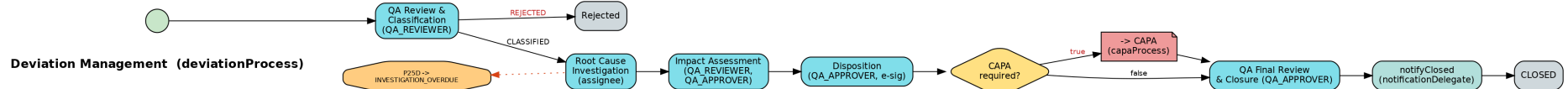
8. Roles & Candidate Groups

Role / group	Responsibility in the workflows
END_USER	Initiates records and performs owner/assignee tasks (investigation, execution, response).
QA_REVIEWER	Reviews and classifies; impact assessment, effectiveness check, response review.
QA_APPROVER	E-signed disposition, approval and closure decisions.
OWNER	Named record owner / assignee for investigation and action steps.
AUDITOR / LEAD AUDITOR	Conducts audits and verifies finding closure.
TRAINING_COORDINATOR	Assigns and verifies training; receives training escalations.
DOC_REVIEWER	Reviews document drafts in the document workflow.
RA_REVIEWER / APPROVER	Reviews regulatory filings on change requests.
CALIBRATION_Tech	Performs equipment calibration and re-calibration.

9. Module-by-Module Workflows

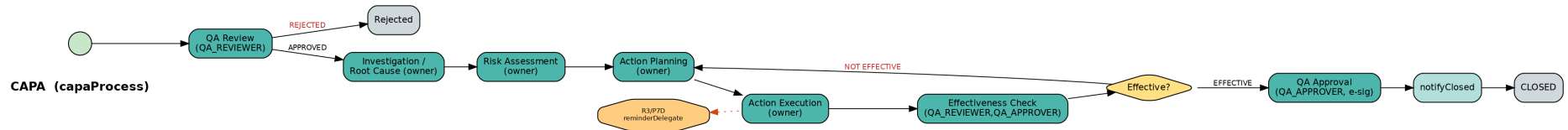
Compact activity flow for each module. Pink/blue/grey rounded boxes are Flowable user tasks (with role); yellow diamonds are gateways; green/teal boxes are service tasks; orange octagons are timers; red notes are cross-module triggers into another process.

1. Deviation Management (deviationProcess)



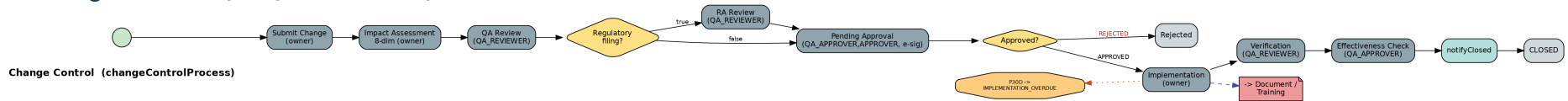
Captures unplanned departures from approved procedures. QA reviews and classifies (Critical/Major/Minor), an owner investigates root cause, QA performs a 4-dimension impact assessment and an e-signed disposition, and — if required — initiates a CAPA before QA closure.

2. CAPA (capaProcess)



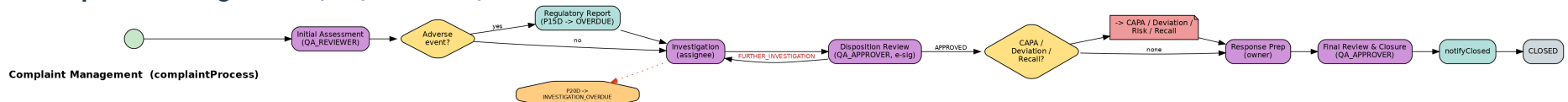
The corrective/preventive-action engine and central integration hub. Almost every other module can spawn a CAPA. Flow: QA review → investigation/root cause → risk assessment → action planning → execution (with reminder timers) → effectiveness check (loops back if not effective) → e-signed QA approval and closure.

3. Change Control (changeControlProcess)



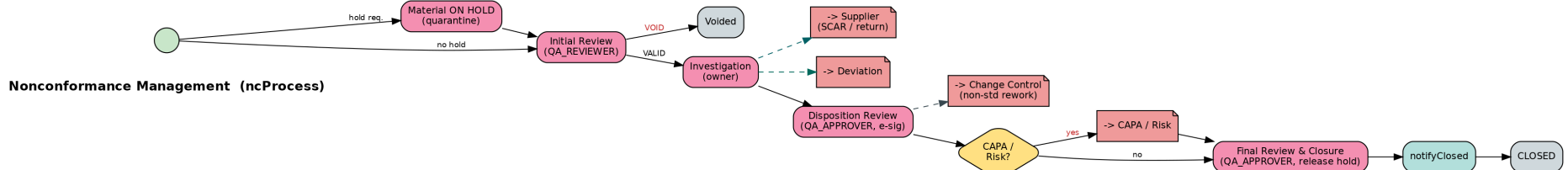
Governs planned changes. Owner submits and runs an 8-dimension impact assessment, QA reviews (optional RA review when a regulatory filing is required), a multi-party e-signed approval gates implementation, then verification and effectiveness check. Implementation carries a 30-day overdue escalation timer.

4. Complaint Management (complaintProcess)



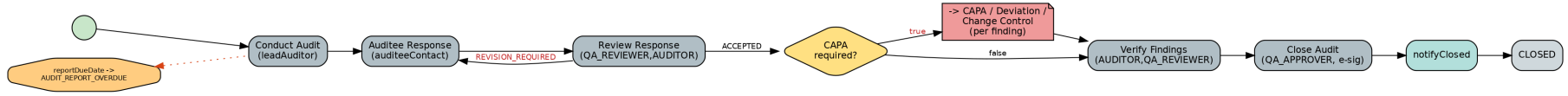
Handles customer/product complaints. Initial assessment classifies severity, adverse-event and regulatory reportability; adverse events fork to a parallel 15-day regulatory report; investigation, e-signed disposition, optional CAPA/Deviation/Recall, formal response and QA closure.

5. Nonconformance Management (ncProcess)



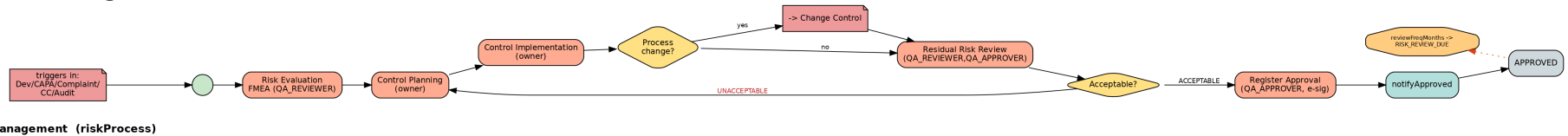
Manages nonconforming material with optional immediate quarantine hold. Initial review (Valid/Void), investigation, supplier SCAR where applicable, e-signed disposition (Use-as-is / Rework / Reprocess / Return / Scrap / Reject), optional CAPA and risk update, then closure with hold release.

6. Audit Management (auditProcess)



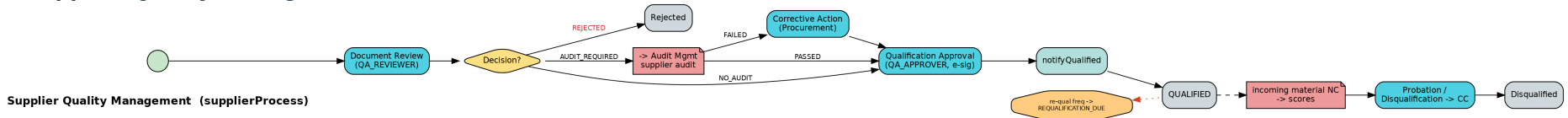
Plans and executes internal/external/supplier audits. Lead auditor conducts the audit (report-due escalation timer), auditee responds, QA/auditor review, findings can initiate CAPA, Deviation and/or Change Control, findings are verified, then e-signed audit closure.

7. Risk Management (riskProcess)



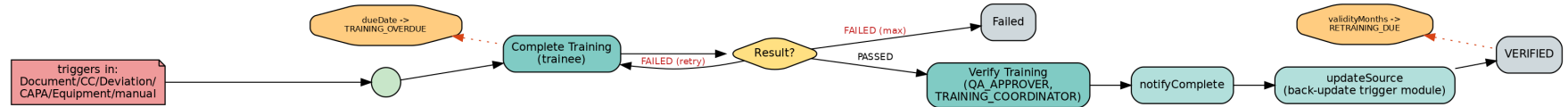
Maintains risk registers and assessments (FMEA/HACCP/FTA/PHA). Receives risk triggers from Deviation, CAPA, Complaint, Change and Audit; evaluates and scores, plans and implements controls (may raise a Change), re-scores residual risk, and e-signs register approval with periodic-review reminders.

8. Supplier Quality Management (supplierProcess)



Qualifies and monitors suppliers. Document review, optional supplier audit (scheduled via Audit Management), corrective action on failure, e-signed qualification. Ongoing monitoring drives re-qualification audits, incoming-material nonconformances, probation and — on disqualification — an alternate-sourcing Change Request.

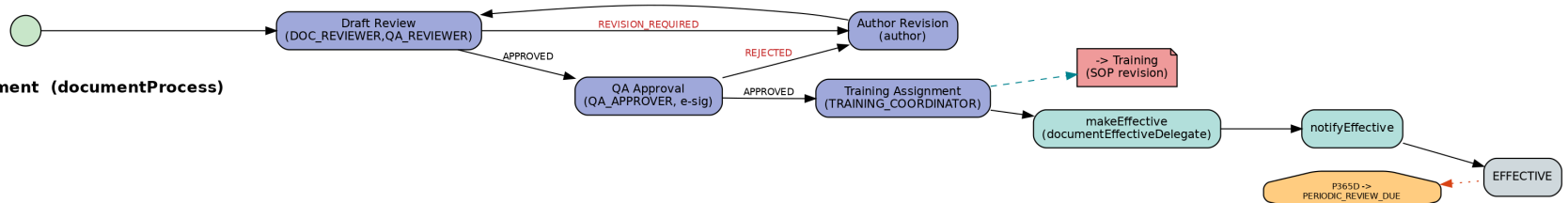
9. Training Management (trainingProcess)



Training Management (trainingProcess)

Assigns and verifies training. Triggered by effective documents, change implementation, deviation/CAPA actions, equipment qualification or manual assignment. Trainee completes (overdue escalation, retry-on-fail), QA/coordinator verify effectiveness, then the source record is back-updated; validity timers drive retraining.

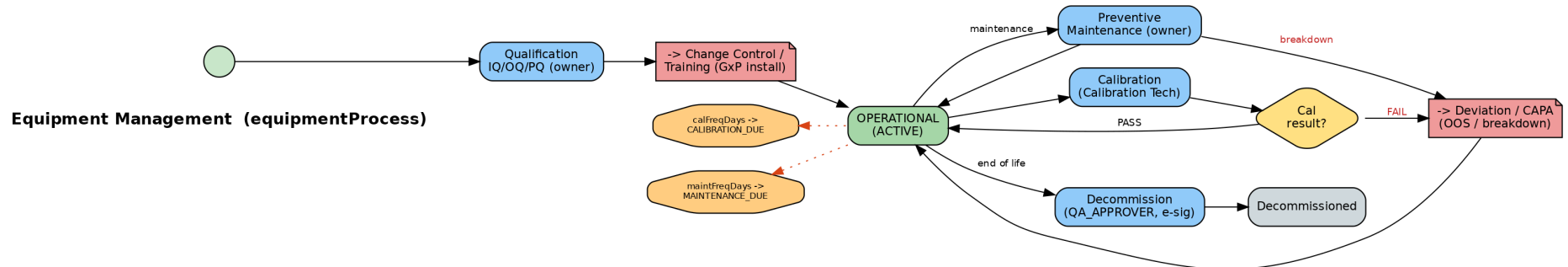
10. Document Management (documentProcess)



Document Management (documentProcess)

Controls SOPs and quality documents. Draft review with author-revision loop, e-signed QA approval, training assignment, then the document is made effective (documentEffectiveDelegate) and stakeholders notified; a 365-day timer drives periodic review.

11. Equipment Management (equipmentProcess)



Manages equipment lifecycle. IQ/OQ/PQ qualification (GxP installs raise Change Control and operator Training), then recurring calibration and preventive-maintenance cycles with due-reminders; calibration failure or breakdown raises a Deviation (and possibly CAPA); periodic re-qualification and e-signed decommission close the lifecycle.

10. Summary

The eleven Mlabs QMS modules are integrated not by point-to-point code but by a shared workflow fabric. Each module is an independent Spring Boot service with its own BPMN process, yet all run on one Flowable engine, post to one unified task inbox, raise notifications and escalations through one notification service, and persist to one PostgreSQL schema with cross-module foreign keys. The connecting points are explicit Flowable service-task delegates that create linked, traceable records and launch the downstream process — with CAPA as the central corrective hub and Risk, Change Control and Training as the principal governance and propagation hubs. The result is a closed-loop quality system: an event detected in one module (a complaint, a nonconformance, an audit finding, an equipment failure) flows automatically into investigation, corrective action, risk, change, document and training processes, each fully audited and electronically signed.

Companion file: `QMS-INTEGRATED-workflow-diagram.mmd` (editable Mermaid version of the master integration map) has been saved alongside this PDF in the docs folder.